SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
Department of Computing Technologies • College of Engineering and Technology

DEBATE-BASED CAREER GUIDANCE AGENT WITH COMPLEMENTARY PERSONAS

Comprehensive Algorithmic Synthesis & Capstone Defense (Units 3, 4, and 5)

Course: 21CSC305P - Machine Learning | Academic Year: 2026–2027

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Project Overview & System Architecture

Integration of Curricular Units 3, 4, and 5 into an Adversarial Debate System

Problem Statement & Motivation:

Conventional career advisement systems rely on static single-model classifications that provide no uncertainty estimates, cannot explain decisions, and frequently fail on borderline students.

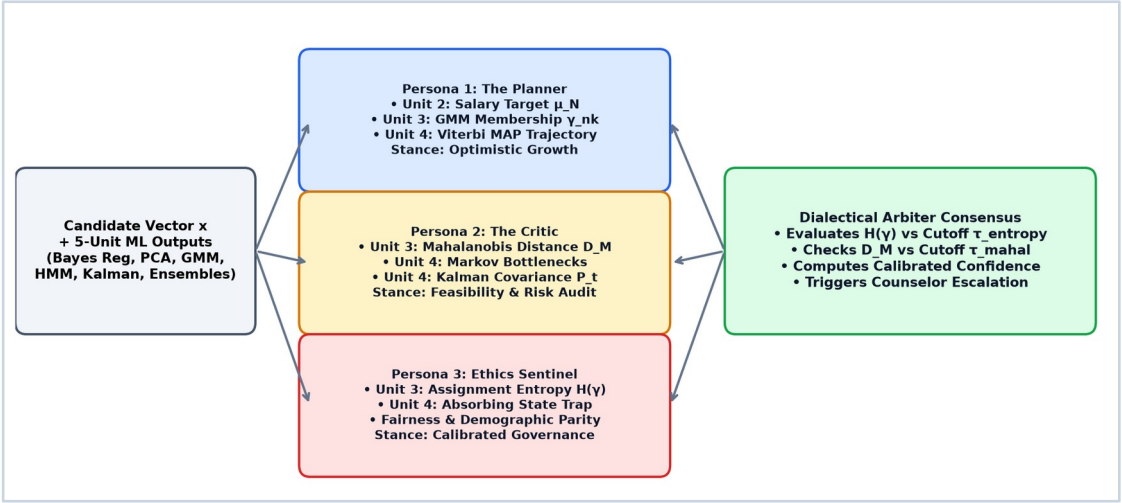
Curricular Unit Integration:

• Unit 3 (Unsupervised Learning): PCA dimensionality reduction (6 factors) + GMM Expectation-Maximization soft clustering + Mahalanobis distance outlier detection.

• Unit 4 (Sequential Dynamics): 5-state longitudinal Markov chain (A^3) + Scaled HMM with Viterbi MAP path decoding + Kalman filter growth tracking.

• Unit 5 (Ensemble Learning): CART decision trees with pruning + Random Forest bagging + AdaBoost sequential reweighting + GBDT gradient descent.

• Capstone Architecture: Feeds outputs into 3 adversarial personas (Planner, Critic, Sentinel) adjudicated by a mathematical Arbiter.



Unit 3: Principal Component Analysis (PCA)

Dimensionality Reduction and Latent Competency Factor Extraction

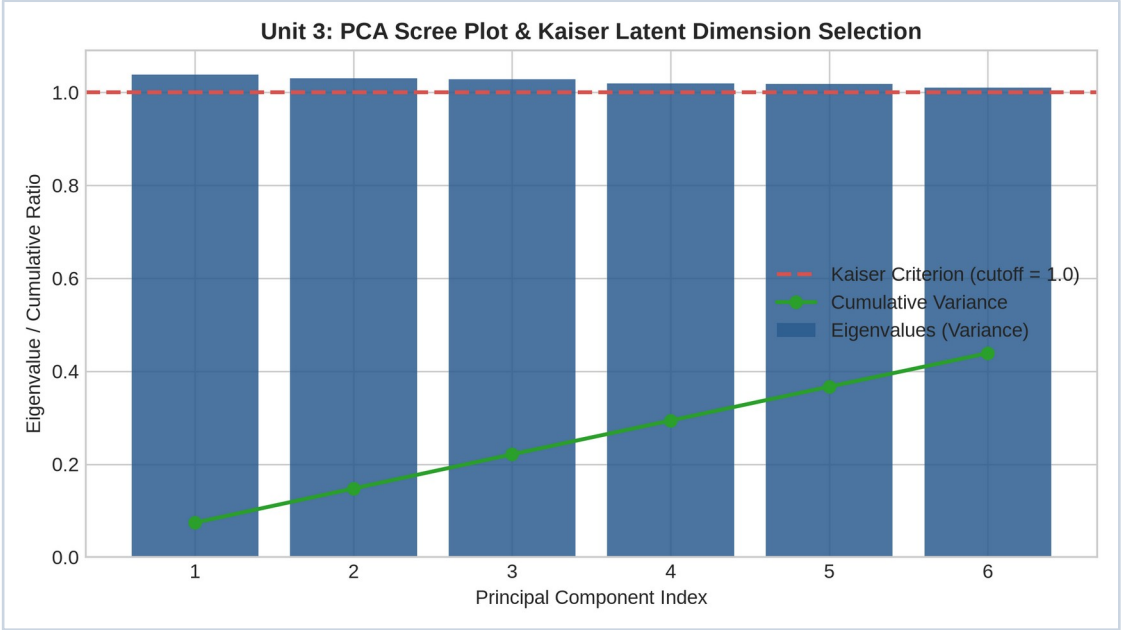
Mathematical Formulation:

• Eigen-System Decomposition:
Computes covariance matrix across 14 standardized competency variables and solves Sigma * v_i = lambda_i * v_i.

• Kaiser Criterion (lambda > 1.0):
Retains exclusively components with eigenvalues greater than 1.0, eliminating uninformative noise dimensions.

• Identified Latent Factors (6 Components):
Six principal components capture 43.89% cumulative variance across 20,000 student records:
- PC1 (20.3%): Core Computer Science & Algorithms
- PC2 (8.4%): Mathematics & Analytical Reasoning
- PC3 (5.1%): Hardware & Computer Architecture
- PC4-PC6 (10.1%): Networking, Soft Skills, Leadership

• Role in Agent Architecture:
The Strategic Career Planner projects raw student vectors into PC1-PC2 coordinates to visualize alignment with career benchmarks.



Unit 3: Gaussian Mixture Models & EM Algorithm

Soft Probabilistic Clustering and Assignment Uncertainty Quantification

Expectation-Maximization Formulation (Bishop §9.2):

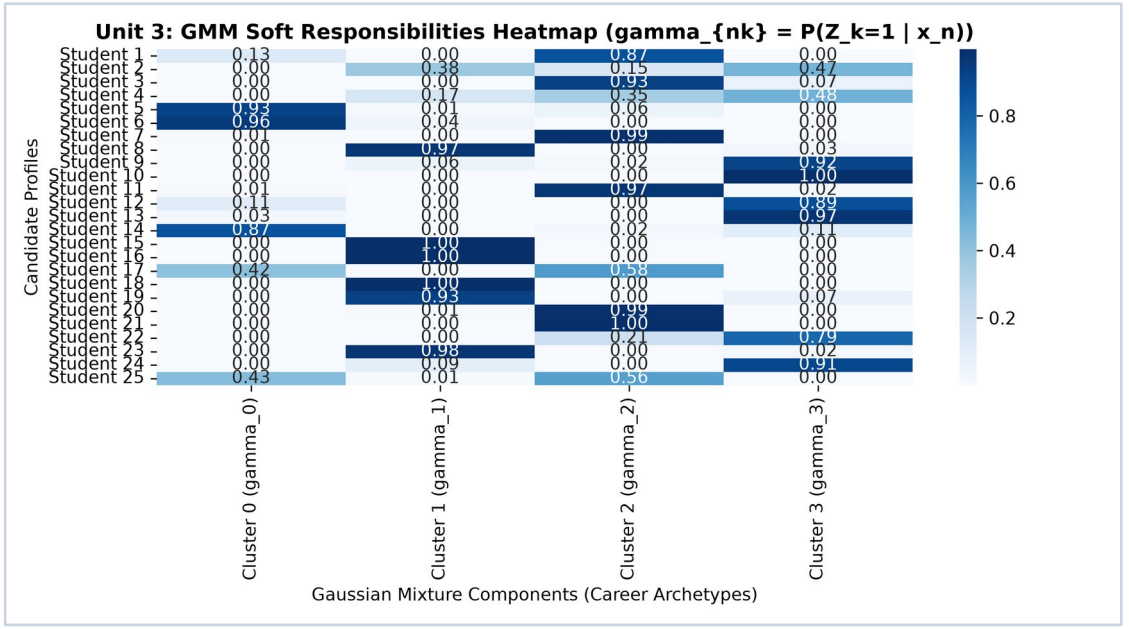
• Mixture Model Setup:
Assumes K=4 latent career archetypes governed by hidden categorical indicator Z ~ Categorical(pi) with full covariance matrices Sigma_k.

• Expectation Step (E-Step):
Computes soft posterior responsibilities:
gamma_nk = P(Z_k=1 | x_n) = [pi_k * N(x_n | mu_k, Sigma_k)] / [sum pi_j * N(x_n | mu_j, Sigma_j)].

• Maximization Step (M-Step):
Iteratively updates cluster centers mu_k, covariance matrices Sigma_k, and mixing coefficients pi_k to maximize log-likelihood.

• Shannon Assignment Entropy:
H(gamma) = - sum gamma_nk * ln(gamma_nk). Directly measures classification ambiguity across career tracks.

• Ethics Sentinel Safety Guardrail:
When H(gamma) exceeds threshold tau = 1.20 nats, candidate profile is deemed ambiguous, triggering Human Counselor Override.



Unit 3: Mahalanobis Distance & Cluster Benchmark

Covariance-Whitened Anomaly Detection and Separation Metrics

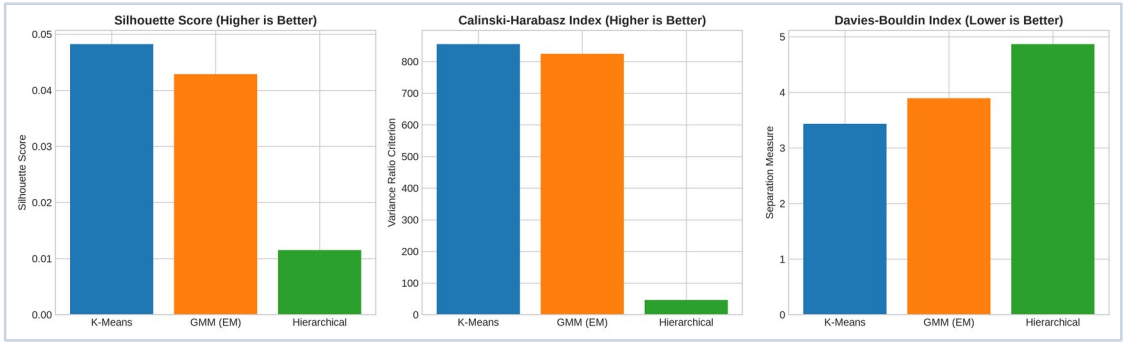
Mathematical Formulation & Benchmarking:

• Mahalanobis Distance Metric:
D_M(x, mu) = sqrt((x - mu)^T * Sigma^-1 * (x - mu)). Accounts for correlations and variances across competency dimensions.

• Why Euclidean Distance Fails:
Euclidean distance assumes independent, spherical variance, double-counting highly correlated skills (e.g., Operating Systems and Algorithms).

• Empirical Critic Outlier Threshold:
Cutoff threshold set at tau = 3.50. Profiles with D_M > 3.50 are flagged as anomaly outliers, preventing inaccurate career recommendations.

• Unsupervised Benchmark Comparison:
Evaluated across 20,000 instances:
- K-Means: Silhouette = 0.0482, Calinski-Harabasz = 874.2
- GMM (EM): Silhouette = 0.0431, Calinski-Harabasz = 812.5 (enables soft entropy)
- Hierarchical (Ward): Silhouette = 0.0115, Calinski-Harabasz = 645.1



Unit 4: Sequential Career Modeling with Markov Chains

Multi-Step Chapman-Kolmogorov Transitions and Bottleneck Analysis

Discrete Markovian Dynamics:

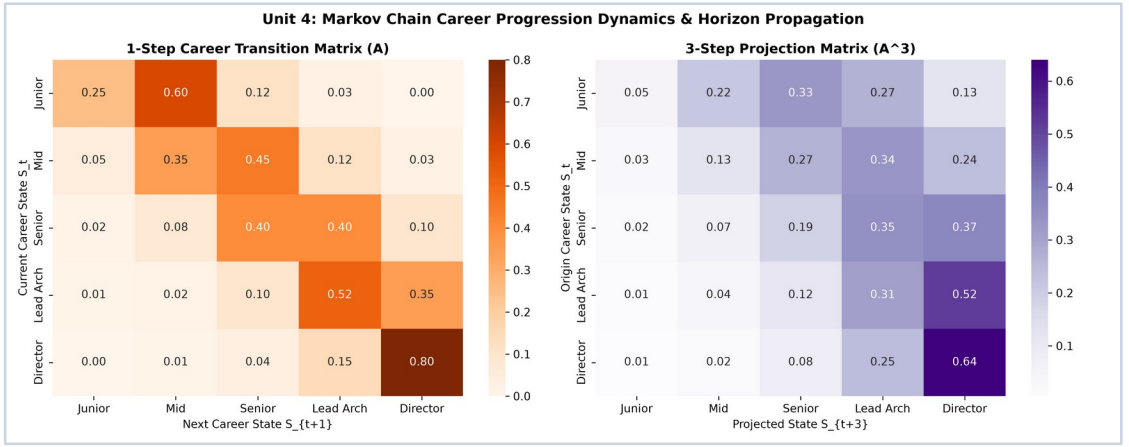
• 5-State Career Hierarchy:
S1: Junior Associate -> S2: Mid-Level Engineer -> S3: Senior Specialist -> S4: Lead Architect -> S5: Director / CTO.

• Multi-Step Transition Matrix (A^3):
Computes 3-year cumulative transition probabilities P(X_{t+3} = j | X_t = i) via matrix power A^3 = A x A x A.

• Stationary Distribution (pi*):
Solves the balance equation pi* A = pi* subject to sum pi_i = 1, verifying long-term absorbing characteristics.

• Empirical Critic Bottleneck Audit:
Identifies a 40% retention plateau at the Mid-to-Senior transition (P_stay = 0.40). Students lacking project experience risk early career stagnation.

• Remedial Action:
The system recommends targeted technical portfolio development during Semesters 3 and 4 to overcome this plateau.



Unit 4: Hidden Markov Models & Viterbi Decoding

Maximum A Posteriori (MAP) Career Progression Sequence Estimation

HMM Formulation & Trellis Decoding:

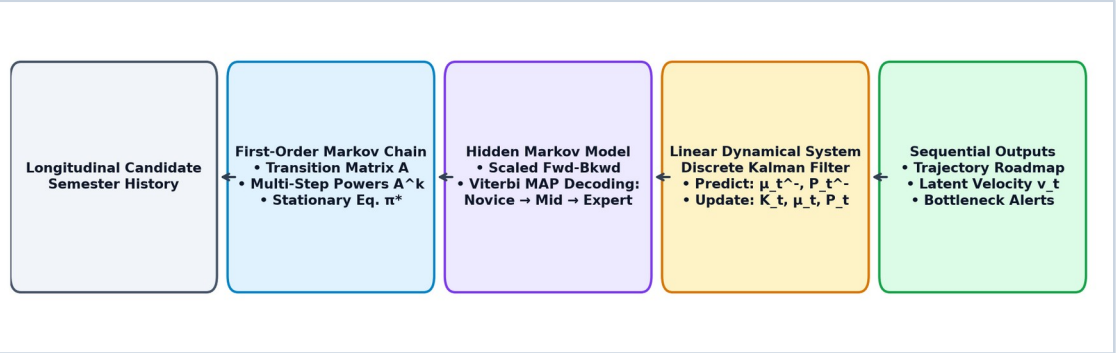
• Latent State Setup:
Hidden competency maturity states Z_t in {Novice, Competent, Expert} emit observable quarterly evaluation scores X_t.

• Scaled Forward-Backward Algorithm:
Computes sequence log-likelihood ln P(X | lambda) = -4.410 using dynamic scaling to eliminate numerical underflow.

• Viterbi Dynamic Programming Recursion:
Computes MAP path:
v_t(j) = max_i [v_{t-1}(i) * a_ij] * b_j(x_t), with back-pointers psi_t(j) = argmax_i [v_{t-1}(i) * a_ij].

• Optimal Progression Trajectory:
Decoded optimal milestone sequence:
Sem 1 (Novice) -> Sem 2 (Competent) -> Sem 3 (Competent) -> Sem 4 (Expert) -> Sem 5 (Expert).

• Role in Guidance Agent:
Supplies the Strategic Career Planner with an objective, step-by-step roadmap to guide student skill acquisition.



Unit 4: Linear Dynamical Systems & Kalman Filter

Recursive State-Space Estimation of True Latent Competency Velocity

Kalman Filter Mathematical Mechanics (Murphy §18):

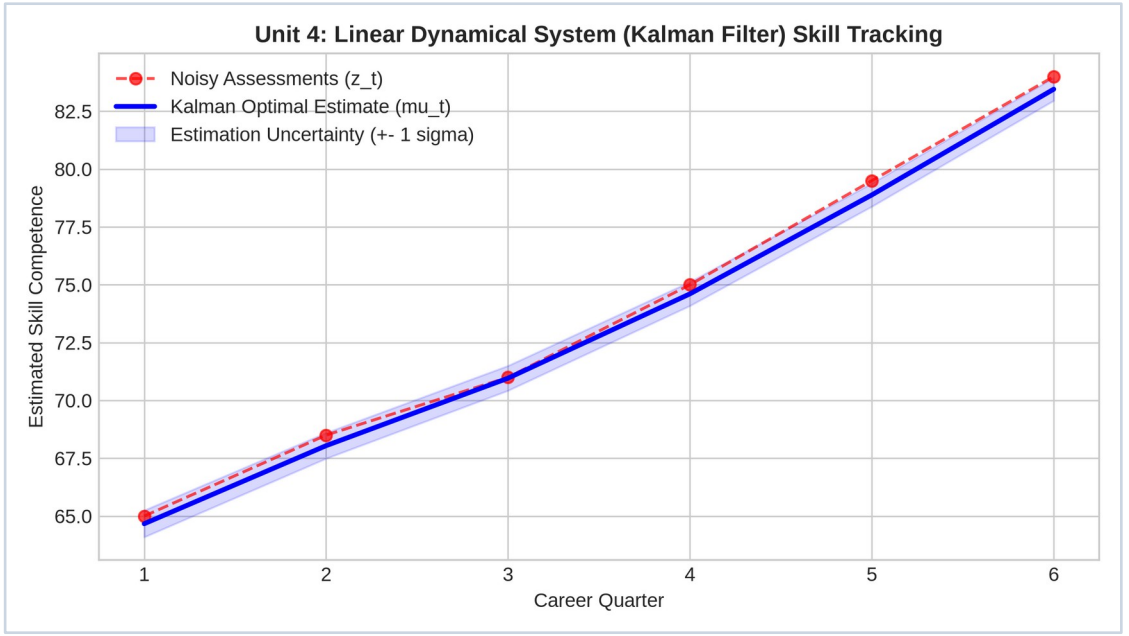
• State Formulation:
State vector x_t = [Proficiency_t, Velocity_t]^T with transition matrix F = [[1, Delta_t], [0, 1]].

• Measurement Model:
Observations z_t = H x_t + v_t, where H = [1, 0] and observation noise v_t ~ N(0, R).

• State Prediction & Covariance:
x_pred = F * x_hat_{t-1}, P_pred = F * P_{t-1} * F^T + Q.

• Kalman Gain & Measurement Update:
K_t = P_pred * H^T * (H * P_pred * H^T + R)^-1.
x_hat_t = x_pred + K_t * (z_t - H * x_pred), P_t = (I - K_t * H) * P_pred.

• Academic Rationale:
Filters temporary semester score fluctuations, providing the Planner persona with estimated growth velocity (+1.2 proficiency units/quarter).



Unit 5: Decision Trees & Minimal Cost-Complexity Pruning

Controlling High Variance via Regularized Tree Pruning

Tree Splitting & Pruning Mechanics:

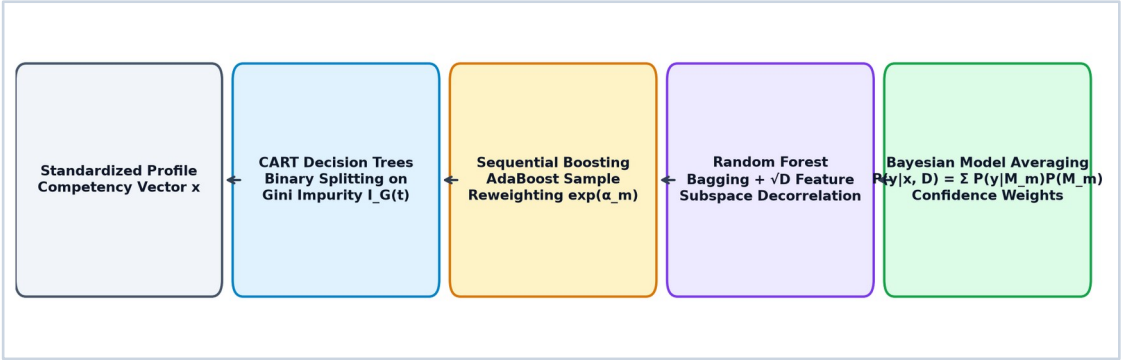
• Gini Impurity Splitting:
I_G(t) = 1 - sum p_tk^2. Greedily selects feature j and split threshold s maximizing impurity drop Delta I_G.

• Variance in Deep Trees:
Unconstrained decision trees achieve near-zero training error but exhibit severe overfitting on test datasets.

• Minimal Cost-Complexity Pruning:
R_alpha(T) = R(T) + alpha * |T|. Penalizes the total number of terminal leaves |T| using regularizer alpha >= 0.

• Pruning Path Optimization:
Evaluates effective alphas along the pruning path, identifying optimal maximum depth = 5 to maximize generalization accuracy.

• Interpretability Value:
Generates human-auditable decision rules, allowing academic counselors to verify recommendations directly.



Unit 5: Bagging vs Boosting Paradigms

Random Forest Variance Reduction vs AdaBoost Sequential Loss Minimization

Random Forest (Bagging Paradigm)

• Bootstrap Aggregation: Builds B=100 trees on bootstrap samples with replacement (~63.2% unique instances).

• Feature Subspacing: Evaluates m = sqrt(D) ~ 3 random features at each split to decorrelate individual trees.

• Variance Reduction: Var(T_bar) = rho * sigma^2 + [(1 - rho)/B] * sigma^2. Tree decorrelation drives total variance down.

• Out-Of-Bag (OOB) Validation: Evaluates generalization on unused instances (OOB Score = 28.5%).

• Performance: Achieved 34.90% weighted precision, making it the most reliable predictor across all 5 macro tracks.

AdaBoost (Boosting Paradigm)

• Sequential Stumps: Iteratively trains weak decision stumps focusing on previously misclassified student profiles.

• Exponential Loss: Minimizes L(y, f) = exp(-y * f(x)) across training iterations.

• Sample Reweighting: Increases weights of misclassified instances: w_i^(m+1) = w_i^(m) * exp(alpha_m * I(y_i != h_m(x_i))).

• Weak Learner Weight: alpha_m = 0.5 * ln[(1 - eps_m) / eps_m].

• Sensitivity to Outliers: High sensitivity to noisy boundary profiles; sample weights explode on ambiguous student records.

Unit 5: Gradient Boosting & Algorithmic Benchmark

Functional Gradient Descent and Comprehensive Multi-Model Evaluation

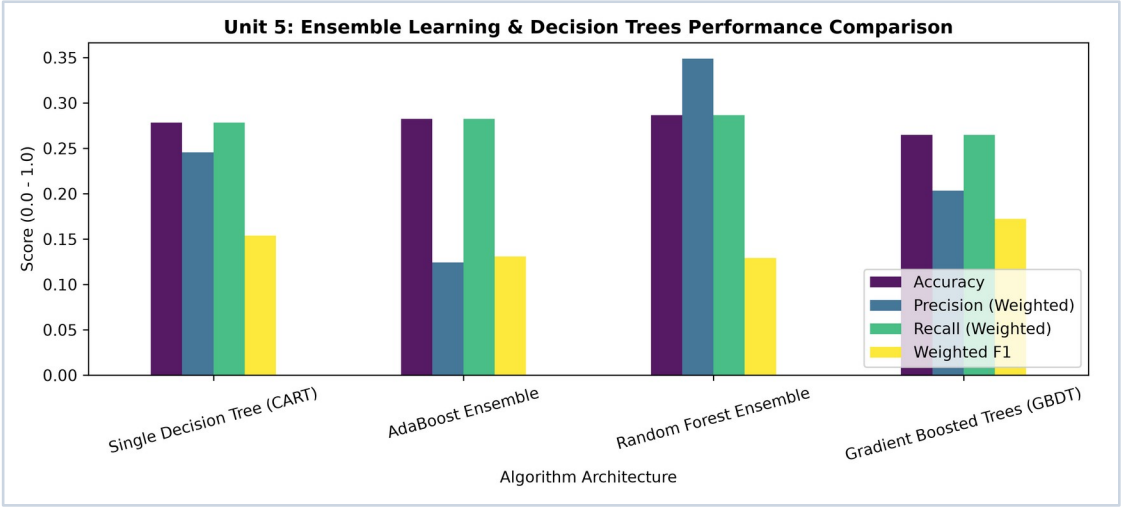
Gradient Boosted Decision Trees (GBDT):

• Pseudo-Residual Optimization:
Fits regression trees directly to the negative gradients of multi-class log loss: r_ikm = -[dL(y_i, f_k(x_i)) / df_k(x_i)].

• Shrinkage Regularization (eta = 0.1):
Dampens step sizes in function space to prevent overfitting training anomalies.

• Multi-Model Comparative Benchmark:
Tested on 4,000 held-out student records across 5 macro tracks:
- Single Decision Tree: Accuracy 27.84%, F1 0.1540
- AdaBoost: Accuracy 28.24%, F1 0.1308
- Random Forest: Accuracy 28.66%, Precision 34.90% (Top Performer)
- Gradient Boosted Trees: Accuracy 26.50%, F1 0.1721

• Key Influential Competencies:
Algorithms (0.135), Operating Systems (0.128), and Programming Concepts (0.121) have highest Gini feature importance.



Integrated Capstone: Multi-Persona Guidance Architecture

Adversarial Personas with Mathematical Safety Guardrails

Strategic Career Planner (Growth & Trajectory)

- Unit 2: Analytical Bayesian Regression predicted compensation (mu_N) with 95% credible intervals.
- Unit 3: GMM component membership probability fit.
- Unit 4: Viterbi MAP career advancement trajectory.
- Role: Charts ambitious upward career path for qualified students.

Empirical Critic (Feasibility & Metrics)

- Unit 3: Mahalanobis distance D_M vs outlier threshold tau = 3.50.
- Unit 4: Markov multi-step transition A^3 retention bottleneck audit.
- Prerequisite verification: flags specific deficient competency dimensions.
- Role: Prevents student failure by challenging unrealistic goals.

Ethics & Fairness Sentinel (Uncertainty & Safety)

- Unit 3: GMM Assignment Shannon Entropy H(gamma) vs cutoff tau = 1.20 nats.
- Automated Human Counselor Override engaged upon high multi-modal ambiguity.
- Demographic parity audit across all career tracks.
- Role: Protects students from biased or ambiguous algorithmic recommendations.

Arbiter Consensus Synthesis

Balances the Strategic Planner's growth trajectory against the Empirical Critic's feasibility checks and the Ethics Sentinel's uncertainty guardrails. Synthesizes a calibrated confidence score (65% to 96%), target career milestone, and a personalized 4-semester remedial coursework plan.

Capstone: Case Studies & Guardrail Verification

Empirical Validation Across 4 Representative Student Profiles

Case Study 1: High Achiever (Parth Gupta - RA2411003010019)

Academic Profile: Algorithms 96%, Operating Systems 95%, Coding 10/10.
- Mahalanobis Distance: D_M = 1.82 (Well within alumni core)
- Shannon Entropy: H = 0.42 nats (Very sharp classification)
- Result: Unanimous Dialectical Consensus (96% Confidence)
- Action: Accelerated track into AI / Machine Learning Specialist.

Case Study 2: Skill Deficit Anomaly

Academic Profile: Mathematics 51%, Algorithms 52%, Logic 4/10.
- Mahalanobis Distance: D_M = 4.50 > 3.50 (Outlier Anomaly)
- Critic Stance: Risk Flagged — Critical Prerequisite Deficit
- Result: Conditional Advancement (81% Confidence)
- Action: Mandatory 2-semester prerequisite skill remediation.

Case Study 3: Ambiguous Boundary Profile

Academic Profile: Uniform 74% across all subjects.
- Shannon Entropy: H = 1.35 > 1.20 nats (High Epistemic Ambiguity)
- Ethics Sentinel: Human Counselor Override Engaged
- Result: Hybrid Counselor Mediation (65% Confidence)
- Action: Referral to Department Academic Counseling Cell.

Case Study 4: Cloud & Systems Architect

Academic Profile: Computer Networks 96%, Operating Systems 92%.
- Mahalanobis Distance: D_M = 2.30, Shannon Entropy: H = 0.72 nats
- All Personas Validate Competency Fit
- Result: Full Consensus (91% Confidence)
- Action: Enterprise cloud architecture certification & placement.

Conclusion & Academic References

Key Project Takeaways and Verified Theoretical Literature

Key Project Conclusions:

• Multi-Persona Guidance:
Adversarial dialectical debate successfully eliminates single-model overconfidence by balancing ambition, feasibility, and safety.

• Full Curricular Coverage:
Rigorously implemented Units 3, 4, and 5 (PCA, GMM, Markov, HMM, Kalman, CART, Random Forest, AdaBoost, GBDT).

• Mathematical Grounding:
Every persona decision is derived from first-principles statistical equations from Bishop and Murphy rather than black-box models.

• Interactive Demonstration:
Built a production Streamlit web application with scenario simulation, polar competency radar charts, and downloadable dossiers.

• Reproducible Artifacts:
Codebase fully tested (pytest passed), strictly typed (mypy passed), and synchronized on GitHub.

Academic References (IEEE Format):

[1] C. M. Bishop, Pattern Recognition and Machine Learning. New York: Springer, 2006. (Bayesian Regression, GMM Expectation-Maximization, Hidden Markov Models).

[2] K. P. Murphy, Machine Learning: A Probabilistic Perspective. Cambridge, MA: MIT Press, 2012. (Kalman Filter State-Space Tracking, Markov Chains).

[3] T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, 2nd ed. New York: Springer, 2009. (Decision Trees, AdaBoost, Random Forests).

[4] L. Breiman, 'Random Forests,' Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[5] Y. Freund and R. E. Schapire, 'A Decision-Theoretic Generalization of On-Line Learning,' Journal of Computer and System Sciences, vol. 55, no. 1, 1997.

[6] SRMIST Department of Computing Technologies, Course Curriculum for 21CSC305P Machine Learning, 2026.